TIC 458507124

Eclipsing binary candidate dossier - Star Vetter

CLEAN

```
BLS EPHEMERIS
  Period        2.694652 d
  Epoch (t0)    1570.4505 BTJD
  Duration      0.1700 d
  Depth         2.659 %  (26,593 ppm)
  SNR           47.8

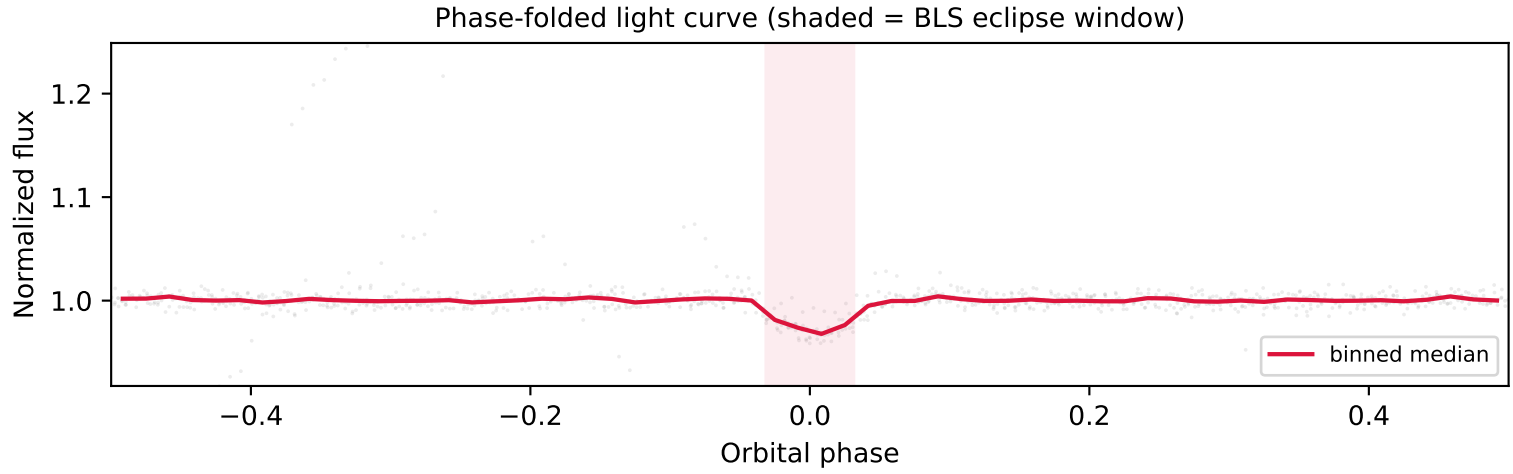
INDEPENDENT CHECKS (f3i gates)
  Eclipses observed     9  (gate >= 3)
  Robust depth ratio    0.99  (gate >= 0.50)
  Robust depth sigma    31.7  (gate >= 7.0)
  Odd/even z-score      1.13  (consistent)
  Secondary eclipse     not detected
```

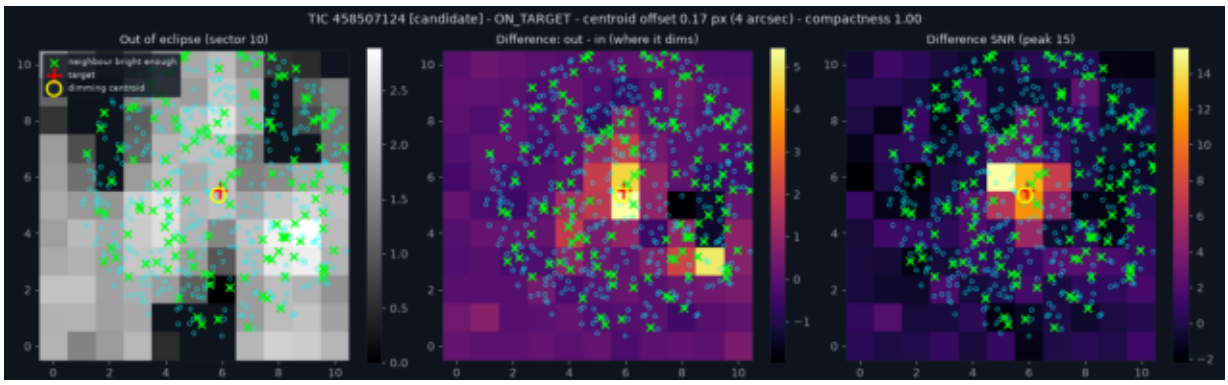
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CATALOG CROSS-MATCH (novelty)

  VSX                                 no match
  ASAS-SN                             no match
  Gaia eclipsing-binary table         no match
  Gaia variable classification        no match
  TESS EB catalog (Prsa+2022)         no match
  ExoFOP TOI list                     no match

  Verdict: NOVEL
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PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)
  Sector 10    light-curve dip in this sector: 30.2 sigma    t0 refined by 0.0000 phase
  Difference-image peak SNR   15.45   (gate >= 5.0)      Compactness   1.00   (gate >= 0.50)
  Dimming centroid offset     0.17 px (4 arcsec)   (gate <= 1.0 px, i.e. on the target)
  Nearest bright-enough neighbour in cutout: TIC 458507127
    0.43 px away, dTmag +2.05 - farther from the dimming than the target
```





No caveats - clean detection on every check.